PROJECT 10

INTELLIGENT CREDIT RISK SCORING & AGENTIC LENDING DECISION SUPPORT

From Financial Risk Modeling to Automated Lending Advice

PROJECT OVERVIEW

This project involves the design and implementation of an **AI-driven credit analytics system** that evaluates borrower credit risk and evolves into an agentic AI lending decision support assistant.

In **Milestone 1**, the system applies classical machine learning techniques to historical borrower data to predict creditworthiness, default probability, and loan risk.

In **Milestone 2**, the same system is extended into an agent-based AI application that autonomously reasons about borrower profiles, retrieves regulatory guidelines, and generates structured credit assessment reports.

The project emphasizes real-world financial risk modeling, progressive development from predictive analytics to agentic AI workflows, and public deployment.

CONSTRAINTS & REQUIREMENTS

TEAM SIZE	API BUDGET
3–4 Students	**Free Tier Only**
FRAMEWORK	**HOSTING**
LangGraph (Recommended)	**Mandatory**

APPROVED TECHNOLOGY STACK

LLMS (MILESTONE 2)

- Open-source models
- Free-tier APIs

AGENT FRAMEWORK

- LangGraph (Recommended)
- Chroma / FAISS (RAG)

ML MODELS (MILESTONE 1)

- Logistic Regression
- Random Forest / Decision Trees
- Scikit-Learn (Pipelines)

UI FRAMEWORK

- Streamlit
- Gradio

HOSTING PLATFORM (MANDATORY)

- Hugging Face Spaces
- Streamlit Community Cloud
- Render (Free Tier)

WARNING: Localhost-only demonstrations will not be accepted.

MILESTONE 1: ML-BASED CREDIT RISK SCORING SYSTEM

MID-SEM SUBMISSION

Objective: Design and implement a machine learning-based credit risk scoring system that predicts borrower creditworthiness using historical loan data. Focus on data preprocessing and supervised learning *without LLMs*.

Functional Requirements:

- Accept borrower/loan data as input.
- Perform data preprocessing (Scaling, Encoding).
- Predict credit risk or default probability.
- Display predictions through a user interface.

TECHNICAL REQUIREMENTS (ML)

- **Preprocessing:** Missing values, Categorical encoding.
- **Models:** Logistic Regression, Decision Trees.
- **Evaluation:** Accuracy, ROC-AUC, Confusion Matrix.
- **Features:** Income, Credit History, Employment.

INPUTS & OUTPUTS

- **Input:** Borrower CSV data.

- **Output:** Credit Risk Score/Class.
- **Metrics:** Key risk driver explanation.

MID-SEM DELIVERABLES

- Problem understanding & Lending Use-case.
- Input–output specification.
- System architecture diagram.
- Working local application with UI.
- Model performance evaluation report.

MILESTONE 2: AGENTIC AI LENDING DECISION SUPPORT

END-SEM SUBMISSION

Objective: Extend the scoring system into an agentic AI lending decision support assistant that autonomously reasons about borrower risk, retrieves regulatory guidelines, and generates structured reports.

Functional Requirements:

- Accept lending queries and borrower profiles.
- Analyze credit risk predictions autonomously.
- Retrieve financial regulations/guidelines.
- Generate structured lending recommendations.

TECHNICAL REQUIREMENTS (AGENTIC)

- **Framework:** LangGraph (Workflow & State).
- **RAG:** Retrieval of regulations (Chroma/FAISS).
- **State:** Explicit state management across steps.
- **Prompting:** Strategies to reduce bias/hallucinations.

STRUCTURED OUTPUT

- **Profile:** Borrower summary & Risk Analysis.
- **Decision:** Recommended lending action.
- **Sources:** Regulatory references.
- **Disclaimer:** Legal & Ethical notices.

END-SEM DELIVERABLES

- Publicly deployed application.
- Agent workflow documentation.
- Structured lending assessment report.
- GitHub Repository & Complete Codebase.
- Demo Video (Functionality walkthrough).

Final Artifacts: Hosted Link, GitHub Repo, Demo Video.

EVALUATION CRITERIA

PHASE	WEIGHT	CRITERIA
Mid-Sem (Milestone 1)	25%	• Correct application of ML techniques • Quality of Preprocessing & Feature Engineering • Appropriateness of evaluation metrics • UI Usability & Code modularity
End-Sem (Milestone 2)	30%	• Quality of Agentic Reasoning & Decision Support • Correct RAG integration (Regulations)

PHASE	WEIGHT	CRITERIA
		• Fairness & Responsible AI Practices • Deployment Success